

Working notes: the zonality of SOC turnover

Storyline decision and diagnostic investigation of the prediction maps

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Purpose. Capture the state of the discussion so we can resume cleanly. This documents (i) the chosen storyline, (ii) what we learned trying to build a global “zonality” map, (iii) a diagnostic investigation of the prediction pipeline (including two false alarms and one real issue), and (iv) where the story now stands and the next concrete step.

1 Storyline decision

We commit to **framing A**: the *zonality of SOC decay* as an *interpretive framework*, presented as the main contribution. The argument: SOC persistence is an emergent ecosystem property (Schmidt et al. 2011); if so, transit times (τ) carry a spatially structured residual *beyond climate*, analogous to classical pedogenic zonality. The deliverable that turns this from slogan into tool is an observation-constrained *map* of that beyond-climate modulation.

Supporting roles: the global τ map and CMIP6/ESM contrast are the empirical payload (B); the edaphic/mycorrhizal mechanism is the interpretive texture (C). The ESM comparison is *repositioned* from “validation” to “motivating illustration” (ESMs concentrate τ variation in a polar extremum and lack mid-latitude structure); this removes the dependence of the headline on the depth/pool comparability caveat, since the load-bearing claim becomes how spatial variation is *organised*, not its absolute magnitude.

2 Building the zonality map: what works and what fails

We must define “zone” *from the data*. We tested several constructions and scored each by how much of the **observed climate-residual** of $\ln \tau$ they explain (point level, out-of-bag; η^2 = between-zone variance fraction). Residual variance ≈ 0.32 (on the biological subset).

Construction	explains observed climate-residual	verdict
Continuous RF modulation (M7–M1)	13.4% (ceiling)	validated signal
discretised, 5 zones	10.5%	works
discretised, 8 zones	11.5%	works
argmax “dominant process”	$\sim 0\%$	fails (collinear channels)
single hard-rule tree (CART)	$\sim 0\%$	fails (signal diffuse)
k -means on raw predictors	0%	fails

Table 1: Only a *supervised* construction works: discretising the validated continuous RF “non-climate modulation field” recovers most of the achievable signal at 5–8 zones. A Köppen-style “which process dominates here” map is *not* supported.

Why the intuitive map fails. The three non-climate channels (Edaphic, Land Use, Biological prediction shifts) are spatially collinear: a PCA of the signed $[E, L, B]$ adjustment vector gives PC1 = 67% with near-equal positive loadings (0.59, 0.53, 0.60; pairwise $r = 0.43$ –0.61). The beyond-climate signal is therefore largely a *single shared modulation axis* (“how strongly non-climate processes bend τ away from the climate expectation”), not three independent process-geographies. This is a cleaner, more defensible framing than a process-mosaic — and it matches the data.

Recommended deliverable (3 layers):

1. continuous global *modulation field* = primary zonality-intensity map;
2. a discrete 5–8 zone companion (quantile bins of the field) for communication;
3. per-zone driver characterisation (predictor profiles / ALE) restoring the “which process where” reading as an honest *secondary* layer.

3 Diagnostic investigation of the prediction maps

While preparing to build the map we stress-tested the global prediction pipeline (script 14). Net conclusion: **the maps are sound; the only real problem is coverage**. Two alarms were chased down and *refuted*.

Hypothesis	verdict
M1 climate raster is corrupted (mean 35.6 vs point 21.8yr)	refuted — coverage artifact
SoilGrids / covariate scaling bug in script 14	refuted — all 19 predictors $r \approx 1.00$
Raster ($M7 - M1$) does not track the signal (-0.15)	refuted — in-sample extraction artifact
Microbial layers cover only 29% of land	confirmed — the real issue

Table 2: Outcome of the pipeline stress-test.

- **No M1 scaling bug.** The 35.6 vs 22.3 yr gap between M1 and M7 is a *domain* effect: M1 restricted to M7’s footprint = $23.8 \approx M7 = 22.3$. All six climate input rasters match the training ranges; offsets in central tendency are the temperate *sampling bias* of the training points.
- **No covariate scaling bug.** Per-predictor training-vs-raster check at 5,000 points: all 19 predictors (climate, SoilGrids edaphic with `d_factor`, terrain, microbial) have $r \approx 1.00$, median ratio ≈ 1.00 .
- **The -0.15 “inconsistency” was our own error:** comparing observations to *in-sample* raster predictions at the training points the models were fit on yields a meaningless in-bag residual. The correct out-of-bag comparison gives the non-climate adjustment correlating $+0.37$ with the observed climate-residual. *Any earlier raster-based $\eta^2 \approx 0$ numbers were contaminated by this same in-sample baseline; the point-level OOB results above stand.*

The real issue — coverage. M1 (climate) covers 2,152,154 land pixels; the microbial-dependent models (M4/M6/M7) cover only **631,252 = 29% of land** — the microbial layers (SPUN, Barceló, fungal proportion) are NA over $\sim 71\%$ of land (high latitudes, drylands). So the “global” τ map, the ESM comparison, and the planned zonality field inherit these holes precisely where the ESM-divergence story is strongest (boreal/Arctic). This is a data gap, not a code bug.

4 Option 2: M5 as the global-coverage workhorse

To get genuine global coverage we adopt **M5 (Climate + Edaphic + Land Use)** — no microbial dependency — as the main map, with M7 retained as a biologically-enriched overlay where mi-

crobial data exist. The M5 modulation field $\ln(M5/M1)$ is saved at `outputs/MRT_predictions/zonality_modul` (mean 0.018, SD 0.279, range $[-1.47, 2.01]$).

Model	land coverage	ΔR^2 vs climate (OOB)	explains climate-res
M5 (climate+edaphic+landuse)	49% (all soil-bearing land)	0.017	3.7%
M7 (+ biological)	29%	0.063	13.4%

Table 3: Coverage improves (29%→49%) but the abiotic signal is weak.

Key emergent finding. The *biological* (mycorrhizal) group carries most of the beyond-climate zonality, yet it is the least globally mapped. *The strongest beyond-climate signal is the least mappable.* Reframed positively, this is a result in its own right — soil biology is the dominant non-climate control on SOC turnover, and the absence of global soil-biology maps is the limiting factor. It motivates a concrete call for global microbial mapping and supports a **dual-layer deliverable**: a global *abiotic* zonality field (M5) plus a *biological enrichment* overlay (M7) where data permit.

5 Resolving process classes without a per-pixel “which-where” map

The channel collinearity blocks *one* thing, not everything. Keep two questions apart:

- **Q1** — “**which process class dominates *where*?**” (a per-pixel attribution map). **Blocked.** Channels correlate $r \approx 0.4$ – 0.6 , enough to make a pixel-by-pixel argmax unstable (the salt-and-pepper map). A Köppen-style process-territory map is *not* supportable.
- **Q2** — “**which process *class* carries beyond-climate signal, and how much?**” (aggregate). **Works.**

Key reconciliation: correlated $\neq$ redundant. Correlation ~ 0.5 is enough to break per-pixel argmax (Q1 fails) but *not* enough to make classes redundant: biology still adds large *unique* variance (M7 reaches 13.4% of the residual vs M5’s 3.7%). “Collinear” was too strong — the channels are correlated, not redundant.

Resolve Q2 by nested increments, not 3-way attribution. Use the nested ladder $M1 \rightarrow M5 \rightarrow M7$. Each increment is a *single* comparison, so its magnitude *and* geography are interpretable (unlike a simultaneous 3-way argmax):

- $M5 - M1 =$ **abiotic** modulation (edaphic + land use; land use is redundant, so $\approx$ edaphic). Globally mappable ($\sim 49\%$ /all soil).
- $M7 - M5 =$ **biological enrichment** (the *unique* biological contribution given abiotic + climate). Substantial and largely independent (since $M7 \gg M5$), on the 29% footprint.

Caveat: nested ΔR^2 is order-dependent (“biology given abiotic”); a commonality analysis (shared vs unique) is the rigorous version, but the headline (biology adds much beyond abiotic; land use redundant) is robust.

Coverage two levels = Tier 2. The coverage limit is not a bolt-on caveat; it *is* the scope boundary between the nested tiers: **Tier 1** (a beyond-climate zonality exists and is spatially organised) is the overarching thesis and does *not* depend on the 29% — it is visible in M5 over all soil-bearing land. **Tier 2** resolves that zonality into a global abiotic level ($M5 - M1$) and a biological-enrichment zoom-in ($M7 - M5$, 29%). Narrative: a global Level-1 abiotic zonality, then a Level-2 deep-dive where soil biology is mapped. “Biological conclusions hold within the microbially-mapped domain” + a call for global soil-biology maps.

6 Routes to a clustered zone map

Two distinct jobs: (i) **discover** the zones (which pixels group together) and (ii) **describe** them with rules (portable, interpretable). Three constructions, increasing in richness. *Terminology: call the classes **zones** / **turnover regimes**, not “syndromes”.*

- **Route A** — **cluster the nested-increment space** [$M1, M5-M1, M7-M5$]. Cheap internal sanity check only; uses existing rasters. Collinearity-safe (increments, not raw channels).
- **Route C (alone)** — **regression tree on the modulation field**. One step: leaves are zones, splits are rules. Zones defined by *level* of modulation (intensity bands). Transparent, portable, Köppen-like.
- **Route B** — **cluster per-pixel driver contributions (TreeSHAP)**. Each pixel gets a $\pm$ -years budget per predictor; clustering groups pixels shaped by the same driver combination. Richest for *insight*, but attribution has its own collinearity caveat and is not portable (needs the RF + SHAP to classify a new site). An analysis, not a tool.

The B→C bridge. B and C are not a forced chain — B *discovers*, C *describes*. To connect them, take B’s per-pixel cluster labels and fit a classification tree whose *response is the cluster ID* and whose inputs are the predictors. That tree emits a rule per zone (“if sand > 60% → zone 3; else if fungal > 0.2 & pH < 5 → zone 5”). Why bother over C-alone: a regression tree merges regions with the same τ -shift even if driven differently (a dryland shortened by sand vs by fungal + low-pH both read “−5 yr” → same leaf); B separates them by driver profile, C then names them. So B→C yields *mechanism-differentiated* zones; C-alone yields intensity bands.

When to use which. Build C-alone first (cheap). Only reach for B→C if intensity-only zones visibly lump mechanistically-different regions. If the modulation is ~ 1 -D (PCA PC1 = 67% suggests it may be), C-alone suffices and B adds little. **Recommendation:** lead with **C** as the flagship (portable rule-based zonation); use **B** to explain what each zone means; keep **A** as a throwaway check. Final manuscript object: a zonation map + a table giving each zone its (i) defining rule, (ii) characteristic τ -modulation, (iii) dominant drivers.

7 Caveats that must be resolved before drawing final figures

1. **OOB understates the signal.** The diagnostics above use OOB, where climate $R^2 = 0.53$; the manuscript uses *spatial CV*, where climate $R^2 = 0.28$. Spatial CV leaves a much larger residual for non-climate to explain, so the honest (spatial-CV) effect sizes are *larger* than the 3.7%/13.4% quoted here. **Re-validate the M5/M7 modulation fields with spatial-block CV** (reuse the existing 2° blocks).
2. **Edaphic ΔR^2 discrepancy.** Quick OOB gives edaphic+landuse only +0.017, whereas the manuscript reports edaphic-alone +0.076 (spatial CV). Almost certainly the OOB-vs-spatial-CV difference, but it is central to the variance-partition claims and must be reconciled explicitly.
3. **Map coherence.** At 0.1° the modulation field is spatially noisy, so quantile-bin zones fragment. Coherent (Köppen-like) zones require coarser resolution or spatially-constrained clustering.
4. **The 29% may be escapable.** It is the *intersection* of ~ 9 biological layers. A “biology-light” model carrying only the single strongest biological variable — fungal proportion (Yu 2022, ~ 1 km, likely near-global) — would intersect far less and could push the biological tier toward $\sim 49\%$ while keeping the signal that mattered most in the ALE. **Check per-layer coverage first** (SPUN richness/endemism is the likely bottleneck).
5. **Biological footprint biased *within* the 29%.** Microbial data are dense in temperate Europe/N. America, thin in the tropics and high latitudes; biological conclusions are

therefore doubly scoped.

8 Where we stand, and the next step

Standing: (i) storyline A is chosen and viable; (ii) the prediction pipeline is verified sound; (iii) the global zonality map is achievable as a continuous modulation field + discrete companion, but only *supervised* constructions work; (iv) the abiotic field (M5) is global-but-modest while the strong signal is biological-but-coverage-limited — now framed as a finding via a dual-layer (M5 global + M7 enrichment) deliverable.

Immediate next step (on reopen): set up **spatial-CV validation** of the M5 and M7 modulation fields, which simultaneously (a) yields paper-consistent effect sizes for the zonality signal and (b) reconciles the edaphic ΔR^2 against the manuscript’s +0.076. The magnitude this returns decides whether the zonality story is “modest but real” or “substantial,” i.e. how hard we can lean on framing A.

Then: (a) check per-layer biological coverage (is the 29% escapable via a fungal-proportion-only model?); (b) build the clustered zone map — lead with Route C (rule-based zonation) at a coherent resolution, add Route B (TreeSHAP contributions) to characterise each zone, with Route A as a sanity check; (c) decide whether to pursue microbial gap-filling (Option 1) as a sensitivity analysis to extend the biological tier globally. The deliverable is a global Level-1 abiotic zonality map + a Level-2 biological-enrichment zoom-in, each zone described by rule, τ -modulation, and dominant drivers.

9 Session update (2026-06-26): validation, scale, provenance, and a data-source correction

Spatial-block CV validation (common row set $n = 101,400$, 2° blocks, seed 42, 300 trees). All quantities computed on a single biological-complete row set so nested increments are strictly comparable (unlike the manuscript table, where each model used a different subset). Total $R_{CV}^2 = 0.378$.

- **Order-independent attribution (Shapley, sums exactly to total):** Climate **32.8%**, Edaphic **32.6%**, Biological **24.2%**, LandUse 10.4%. *Climate is only $\sim 1/3$ of the attributable signal; the non-climate domains carry $2/3$.* This is the headline and it is scale-independent.
- **Singles alone:** Edaphic $0.318 \geq$ Climate $0.306 >$ Biological $0.268 \gg$ LandUse 0.129 . Singles sum to 1.021 vs full 0.378 $\Rightarrow$ overlap 0.64 (“correlated, not redundant”). Use *singles-first* as the Results opening.
- **Order-dependence reconciled:** biology added last (CELB–CEL) = +0.008 but Shapley share 24% — same data; LandUse redundant with Edaphic.
- **Modulation field is out-of-sample valid but modest:** $\text{cor}(\text{observed climate-residual, full modulation}) = 0.32$ ($R^2 = 0.10$); discrete 5–8 zones capture $\eta^2 \approx 0.07\text{--}0.09$. Biology is a spatially *separate* axis: $\text{cor}(\text{abiotic modulation, biological enrichment}) = -0.16$.

Scale spectrum and noise floor (appendix material). Residual variogram nugget/sill ≈ 0.66 : $\sim 2/3$ of the beyond-climate residual is unstructured noise (measurement + change-of-support + sub-reliable-resolution covariate error) at >15 km; the variogram is nearly pure-nugget beyond 15 km. η^2/null shows coarse-scale organisation rising from $5\times$ (1°) to $33\times$ (20°)

chance, but the absolute coarse signal is small ($\sim 3\text{--}7\%$ of variance), visible only after block-averaging. Covariates predict the *block-mean* (zonal) residual $2\text{--}3\times$ better than the point residual (R^2 $0.10 \rightarrow 0.20\text{--}0.29$, peaking at $\sim 3\text{--}4^\circ$). *Implication*: build the map at the coarse effective scale; never show native-pixel texture; the scale story is supporting (appendix), not the headline. Figure: `plots/appendix/scale_noisefloor.png`.

Provenance control — PASSES. Tested whether the coarse signal is a source/lab artifact using `source_db` (44 contributing databases). Only **2.4%** of beyond-climate residual variance is between sources; de-meaning by source leaves the coarse organisation almost intact (10° : $24\times \rightarrow 19\times$ chance) and covariate skill barely dented ($0.22 \rightarrow 0.19$ at 5°). The variogram mid-lag bump is unchanged by de-meaning, so it is *not* a between-source offset. Conclusion: the modest zonal signal is **ecological, not provenance**. (Caveat: source is confounded with geography, so this is a conservative/upper-bound test — surviving signal is definitely ecological.)

Data-source correction (important). The SOC data is **not** WoSIS. It is the OpenLandMap *Open Compendium of Soil Datasets: Soil Observations and Measurements* (Zenodo 15593990; `sol_chem.pnts.horizons.rds`). WoSIS is not even a `source_db` label in the data. Manuscript fixed in 4 places (§Soil data compilation, OC=0 sentence, Results data paragraph, Data availability) + new bib entry `OpenLandMapSoilDB2025` (author/year **TODO**: verify against Zenodo record). Source breakdown corrected: post-QC 198,382 profiles from **44** (not 47) databases; `GLanCE.points` (claimed 21.1% largest) is removed entirely by QC — the old paragraph applied the pre-QC breakdown to the post-QC count. Real top sources: USDA_NCSS 20.4%, Netherlands.BIS 11.5%, SISLAC 8.9%, CSIRO_NatSoil 7.0%, AfSPDB 6.7%.

Refined storyline. Lead with *attribution* (climate only $1/3$, robust and scale-free) and the ESM contrast: ESMs get zonality wrong in two independent ways — they *over-amplify* the climate range (polar $50\text{--}100\times$ vs data $\sim 3\text{--}5\times$; magnitude claim, caveated by steady-state/change-of-support) *and* omit the soil+biological axis that carries most of the real organisation (structural claim, robust). Bridge to next-generation models needing explicit mineral+microbial processes. Zonality is *real but modest*; present the map honestly at coarse scale.

10 Open issues / next steps

1. **Promote to production** (must-do before anything enters the MS): house-style `13c_commonality_validation` at 500 trees, writing `outputs/13c*.rds`, the singles $\rightarrow$ pairs $\rightarrow$ full + Shapley figure, the scale/variogram appendix figure, and the provenance-control table. Repeatable via `run_pipeline.R`.
2. **Biological-parsimony appendix (the “29% killer”)**. Check collinearity + per-layer coverage of the 9 biological layers; single-layer marginal ΔR^2 vs coverage; recommend one parsimonious layer to extend the biological footprint well beyond 29%. Figures $\rightarrow$ `plots/appendix/`; TODO comment already placed in `manuscript.tex` before Supplementary tables.
3. **Verify the compendium citation** (authors/year/version) from the Zenodo record; update `OpenLandMapSoilDB2025`.
4. **Add Moran’s I** on block-aggregated residuals as the standard spatial-autocorrelation statistic to accompany the η^2 /null backbone.
5. **Build the dual-layer zonality map** at the coarse effective scale ($\sim 3\text{--}4^\circ$), with per-zone driver characterisation (ALE/TreeSHAP).

6. **Reconcile MS magnitude numbers:** the current “~10 pp beyond climate” is partly a subset artifact; on comparable rows climate = 0.31 and beyond-climate ≈ 7 pp.
7. **Narrative:** write with Schimel OCAR; keep Jungian archetypes as a hidden internal scaffold only (deferred — do not implement in text).

Artifacts (2026-06-25/26): `outputs/MRT_predictions/zonality_modulation.M5.M1.logratio.tif`; `plots/appendix/scale_noise`
exploratory scripts in the session scratchpad (`spatialcv_validation.R`, `spatialcv_finish.R`, `scale_noise`
`provenance_control.R`; earlier `zone_explore.R`, `zone_supervised.R`, `m5_zonality.R`). All reproducible from `outputs/12b.model_rea`
`outputs/13_rf_models.rds`, `outputs/13_var_groups.rds`. *Scratchpad scripts are exploratory and must be promoted*
to house-style pipeline code before any result enters the manuscript.